GREENS TECHNOLOGYS

Greens Technology is a leading Training and Placements company in Chennai. We are known for our practical approach towards trainings that enable students to gain real-time exposure on competitive technologies. Trainings are offered by employees from MNCs to give a real corporate exposure. We make sure that all our sessions are very much interactive and well structured. We preferpractical approach rather than theoretical information to master the technical depth of the technology. We provide excellent Lab Handouts for practice including Real-time Case Studies and Projects on ALL courses. Our Trainings are focused on perfect improvement of technical skills for Fresher's and working professional. Our Training classes are sure to help the trainee with COMPLETE PRACTICAL TRAINING and Real-time methodologies.

AWS CLOUD & DEVOPS TOOLS

COURSE OVERVIEW

AWS & DevOps is a multidisciplinary domain that combines cloud computing, automation, and continuous delivery practices to design, deploy, and manage scalable applications. **AWS (Amazon Web Services)** forms the backbone of cloud infrastructure, offering core services such as EC2, S3, VPC, IAM, RDS, and Lambda to provision compute, storage, networking, and security resources on demand. These services enable organizations to build highly available, fault-tolerant, and cost-optimized systems without managing physical hardware.

DevOps complements AWS by streamlining the software development lifecycle through automation and collaboration. Tools like **Git and GitHub** manage source code and version control, while **CI/CD pipelines using Jenkins or GitHub Actions** automate build, test, and deployment processes. **Infrastructure as Code (IaC)** tools such as **Terraform and AWS CloudFormation** allow teams to define and manage cloud infrastructure programmatically, ensuring consistency and repeatability across environments. Configuration management tools like **Ansible** further automate server setup and application configuration.

Containerization and orchestration play a critical role in modern DevOps workflows. **Docker** packages applications into portable containers, while **Kubernetes and AWS services like EKS and ECS** manage containerized workloads at scale. Monitoring and logging tools such as **Amazon CloudWatch and Prometheus** provide real-time visibility into system performance, enabling proactive issue detection and faster incident response.

Together, **AWS and DevOps** establish a robust ecosystem where cloud infrastructure, automation, and continuous delivery converge. This synergy empowers teams to release software faster, scale effortlessly, maintain high reliability, and adapt quickly to changing business requirements—making AWS & DevOps an essential skill set for modern cloud engineers and site reliability professionals.

No of Days

Topic	No of days	Theory Hours	Practical Hours
AWS	18 days	10 hrs	26 hrs
GIT	3 days	2 hrs	4 hrs
Jenkins	3 days	2 hrs	4 hrs
Docker	3 days	2 hrs	4 hrs
Kubernetes	3 days	2 hrs	4 hrs
Terraform	3 days	2 hrs	4 hrs
Ansible	3 days	2 hrs	4 hrs
Prometheus & Grafana	1 day	1 hr	1 hr
Splunk	1 day	1 hr	1 hr
Linux	1 day	1 hr	1 hr
Python	2 days	1 hrs	3 hrs
Project 1: 3 Tier Architecture on AWS	1 day	1 hr	3 hrs
Project 2: Application Deployment CI CD	1 day	1 hr	3 hrs
Project 3: Infra Deployment CI CD	1 day	1 hr	3 hrs
Total	44 days	88 hrs	

AWS Cloud (36 hrs)

Day 1: Introduction to Cloud

- Introduction to Cloud Computing
- Why Cloud Computing?
- History of Cloud Computing
- Evaluation of Cloud Computing
- Virtualization and Hypervisor
- Cloud Service Models
- Cloud Deployment Models
- Introduction to AWS
- Why Learn AWS?
- Data Centers and Regions
- AWS Global Infrastructure:
- Overview of AWS Regions and
- Availability Zones.
- Understanding the AWS global
- network infrastructure.
- Selecting the appropriate AWs
- Region for deployment.

Day 2: EC2

- What is Amazon EC2?
- EC2 Use Cases
- EC2 Architecture Overview
- Instance Types (General, Compute, Memory, Storage optimized)
- AMI (Amazon Machine Image)
- Key Pairs (Public & Private Keys)
- Security Groups (Inbound & Outbound Rules)
- Launching an EC2 Instance (Hands-on)
- Connecting to EC2 (SSH for Linux, RDP for Windows)
- Understanding EC2 Pricing Models
- On-Demand
- Reserved Instances
- Spot Instances
- EC2 Instance Lifecycle

Day 3: EC2 Advanced

- EC2 Storage Options Overview
- Elastic Network Interfaces (ENI)
- Elastic IP Address
- User Data & Bootstrapping
- EC2 Placement Groups
- Monitoring EC2 with CloudWatch Metrics
- EC2 Auto Recovery
- EC2 Backup using AMI
- Best Practices for EC2
- EC2 Troubleshooting Scenarios

Day 4: EBS Volumes

- What is Amazon EBS?
- EBS Volume Types (gp3, gp2, io1, io2, st1, sc1)
- Creating and Attaching EBS Volumes
- EBS vs Instance Store
- Formatting and Mounting EBS (Hands-on)
- EBS Snapshots
- Incremental Snapshots
- Encrypting EBS Volumes
- Resizing EBS Volume
- EBS Best Practices

Day 5: S3

- Overview of Python and its applications in data analysis Introduction to Amazon S3
- S3 Use Cases
- Buckets and Objects
- S3 Storage Classes
- Versioning
- S3 Permissions (Bucket Policy & ACL)
- S3 Encryption (At-rest & In-transit)
- Static Website Hosting on S3
- Lifecycle Rules
- S3 Replication
- S3 Best Practices

Day 6: IAM

- What is IAM?
- Importance of IAM in AWS Security
- IAM Users
- IAM Groups
- IAM Roles
- IAM Policies (Managed vs Inline)
- Policy Structure (JSON Basics)
- MFA (Multi-Factor Authentication)
- IAM Best Practices
- Shared Responsibility Model

Day 7: VPC

- What is Amazon VPC?
- VPC Components Overview
- CIDR Blocks
- Subnets (Public & Private)
- Internet Gateway
- Route Tables
- Launching EC2 in VPC
- VPC Security Overview

Day 8: VPC Advanced

- NAT Gateway & NAT Instance
- VPC Peering
- VPC Endpoints
- Network ACLs

- Security Groups vs NACLs
- Bastion Host
- VPC Flow Logs
- VPC Architecture Best Practices

Day 9: LB & ASG

- High Availability & Scalability Concepts
- Elastic Load Balancer Overview
- Types of Load Balancers (ALB, NLB, CLB)
- Auto Scaling Group Components
- Launch Templates
- Scaling Policies
- Health Checks
- Hands-on: ALB with ASG

Day 10: RDS

- Introduction to Amazon RDS
- Supported Databases
- RDS Architecture
- Multi-AZ Deployment
- Read Replicas
- RDS Backup & Snapshots
- RDS Security
- Monitoring RDS

Day 11: Cloud Watch, Cloud Trial & SNS

- Amazon CloudWatch (Metrics, Logs, Alarms)
- Custom Metrics
- CloudTrail Overview
- Use Cases of CloudTrail
- Amazon SNS
- SNS Topics & Subscriptions
- Real-time Notifications

Day 12: Cloud Front & WAF

- Introduction to CDN
- Amazon CloudFront
- CloudFront Architecture
- Origins & Behaviors
- Caching & TTL
- AWS WAF Overview
- Protecting Applications using WAF

Day 13: Lambda

- Serverless Computing Concepts
- Introduction to AWS Lambda
- Lambda Architecture
- Creating First Lambda Function
- Lambda Execution Role
- Triggers & Events
- Lambda Pricing
- Best Practices

Day 14: API Gateway & DynamoDB

- Introduction to API Gateway
- REST API vs HTTP API
- Integrating API Gateway with Lambda
- DynamoDB Overview
- Tables, Items & Attributes
- Primary Keys
- Read/Write Capacity
- DynamoDB Use Cases

Day 15: Route 53

- What is DNS?
- Amazon Route 53 Overview
- Hosted Zones
- Record Types
- Routing Policies
- Health Checks
- Domain Registration

Day 16: Code Deploy & Code Pipeline

- CI/CD Overview
- AWS CodeDeploy
- Deployment Strategies
- AWS CodePipeline
- Pipeline Stages
- Integrating GitHub
- End-to-End CI/CD Demo

Day 17: ECS ECR KMS

- Containers Basics
- Amazon ECR
- Docker Image Push & Pull
- Amazon ECS
- Task Definitions
- ECS Launch Types
- AWS KMS
- Encryption Concepts

Day 18: EKS

- Kubernetes Fundamentals
- Amazon EKS Overview
- EKS Architecture
- Worker Nodes
- Pods & Services
- Deploying Application on EKS
- EKS Security Best Practices

GIT (6 hrs)

Day 1

- What is Version Control?
- Centralized vs Distributed Version Control
- Introduction to Git
- Installing Git & Initial Configuration
- Git Repository (Local & Remote)
- Git Workflow Basics
- Common Git Commands
- git init, status, add, commit
- Understanding Git Snapshots
- Hands-on: Create and Manage Local Repository

Day 2

- Git Branches
- Creating & Switching Branches
- Merging Branches
- Merge Conflicts & Resolution
- Introduction to GitHub
- Pushing Code to Remote Repository
- Pull Requests Workflow
- Git Clone, Pull, Fetch
- Hands-on: Team Collaboration using GitHub

Day 3

- Git Rebase vs Merge
- Git Stash
- Git Tags
- Git Reset vs Revert
- .gitignore File
- Git Hooks (Intro)
- Git Best Practices
- Real-world Git Workflow (Feature, Release, Hotfix)

Jenkins (6 hrs)

Day 1

- What is CI/CD?
- Introduction to Jenkins
- Jenkins Architecture
- Installing Jenkins (Linux / Docker)
- Jenkins Dashboard Overview
- Creating Freestyle Jobs
- Source Code Integration (GitHub)
- Build Triggers
- Hands-on: First Jenkins Job

Day 2

- Jenkins Pipeline Concepts
- Declarative vs Scripted Pipeline
- Jenkinsfile Structure
- Pipeline Stages & Steps
- Environment Variables
- Post-build Actions
- Hands-on: CI Pipeline using Jenkinsfile

Day 3

- Jenkins Plugins Overview
- Credential Management
- Parameterized Builds
- Jenkins Agents & Nodes
- Integration with Docker
- Notifications (Email, Slack)
- Jenkins Best Practices
- CI/CD Pipeline Demo

Terraform (6 hrs)

Day 1

- What is Infrastructure as Code?
- Introduction to Terraform
- Terraform Architecture
- Installing Terraform
- Terraform Providers
- Terraform Configuration Files
- terraform init, plan, apply
- Hands-on: Create AWS Resources using Terraform

Day 2

- Terraform State File
- Remote Backend
- State Locking
- Variables & Outputs
- Terraform Modules
- Reusable Infrastructure
- Hands-on: Modular Terraform Setup

Day 3

- Terraform Workspaces
- Dependency Management
- Lifecycle Rules
- Terraform Import
- Best Practices
- Terraform vs CloudFormation
- Real-world IaC Project

Ansible (6 hrs)

Day 1

- What is Configuration Management?
- Introduction to Ansible
- Ansible Architecture
- Agentless Nature of Ansible
- Inventory Files
- Ansible Ad-hoc Commands
- Modules Overview
- Hands-on: Basic Ansible Commands

Day 2

- YAML Basics
- Writing Ansible Playbooks
- Tasks, Handlers, Variables
- Conditionals & Loops
- Roles & Directory Structure
- Hands-on: Server Configuration using Playbooks

Day 3

- Ansible Vault
- Dynamic Inventory
- Error Handling
- Ansible Best Practices
- Ansible Integration with AWS
- Real-world Configuration Management Project

Prometheus & Grafana (2 hrs)

Day 1

- What is Monitoring and Observability?
- Difference between Monitoring, Logging, and Tracing
- Introduction to Prometheus
- Prometheus Architecture
- Time-series database
- Targets & Scraping
- Metrics, Labels, and Time Series
- Prometheus Data Model
- Exporters
- Node Exporter
- Introduction to PromQL
- Basic queries
- Filtering and aggregations
- Introduction to Grafana Architecture
- Connecting Prometheus as a Data Source
- Creating Dashboards and Panels
- Alerts and Alertmanager (Overview)

Splunk (2 hrs)

Day 1

- What is Log Management?
- Why Centralized Logging is Required?
- Introduction to Splunk
- Splunk Use Cases (Monitoring, Security, Troubleshooting)
- Splunk Architecture
    - Forwarder
    - Indexer
    - Search Head
- Universal Forwarder vs Heavy Forwarder
- Data Inputs and Sources
- Indexes and Indexing Process
- Understanding Events and Fields
- Introduction to SPL (Search Processing Language)
- Basic SPL Commands
    - Search
    - Filter
    - Stats
- Creating Reports and Dashboards
- Alerts and Real-time Monitoring
- Hands-on: Log Analysis using Splunk
- Installing Terraform

Python (4 hrs)

Day 1

- Introduction to Python
- Why Python for DevOps & Cloud?
- Installing Python
- Python Syntax and Indentation
- Variables and Naming Conventions
- Data Types Overview
- Operators
    - Arithmetic Operators
    - Comparison Operators
    - Logical Operators
- Conditional Statements
    - if
    - elif
    - else
- Loops
    - for loop
    - while loop
- break and continue statements
- Hands-on: Basic Python Programs

Day 2

- Functions in Python
- Defining and Calling Functions
- Function Arguments
    - Positional
    - Keyword
- Return Statements
- Built-in Functions
- String Data Type
- String Operations
- Common String Functions
- Collections in Python
    - List
    - Tuple
    - Set
    - Dictionary
- Iterating over Collections
- Basic Exception Handling
    - try
    - except
- Hands-on: Python Examples related to DevOps use cases

Docker (6 hrs)

Day 1

- What are Containers?
- Containers vs Virtual Machines
- Introduction to Docker
- Docker Architecture
- Installing Docker
- Docker Images & Containers
- Common Docker Commands
- Hands-on: Run First Container

Day 2

- Dockerfile Basics
- Building Custom Images
- Image Layers & Caching
- Docker Volumes
- Docker Networking
- Exposing Ports
- Hands-on: Containerize a Web Application

Day 3

- Docker Compose
- Multi-container Applications
- Docker Registry & Docker Hub
- Image Tagging & Versioning
- Docker Security Basics
- Docker Best Practices

Kubernetes (6 hrs)

Day 1

- Introduction to Kubernetes Architecture
- Kubernetes Components (API Server, etcd, Scheduler, Controller Manager)
- Understanding Control Plane & Worker Nodes
- Installing Kubernetes using Minikube
- Installing Kubernetes using Kops (AWS-based)
- Comparing Minikube vs Kops
- kubectl Installation & Configuration
- Basic kubectl Commands
- Hands-on: Create and Verify Kubernetes Cluster

Day 2

- What is a Pod?
- Pod Lifecycle
- Multi-container Pods
- ReplicaSet
- Deployment Controller
- Rolling Updates & Rollbacks
- DaemonSet
- Namespace Concepts
- Creating & Managing Namespaces
- Hands-on: Deploy Applications using Deployments

Day 3

- Why Services are Required?
- Service Types Overview
- ClusterIP Service
- NodePort Service
- LoadBalancer Service
- Service Discovery
- DNS in Kubernetes
- Hands-on: Expose Applications using Services

Projects (12 hrs)

- Project 1: 3 Tier Architecture on AWS **(4 hrs)**
- Project 2: Application Deployment CI CD **(4 hrs)**
- Project 3: Infra Deployment CI CD **(4 hrs)**